已镜像和未镜像聚合

ONTAP 9

NetApp
April 11, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/disks-aggregates/mirrored-unmirrored-aggregates-concept.html on April 11, 2022. Always check docs.netapp.com for the latest.

目录

已镜像和未镜像聚合 1
已镜像和未镜像聚合 1
未镜像聚合的工作原理 1
镜像聚合的工作原理 2

已镜像和未镜像聚合

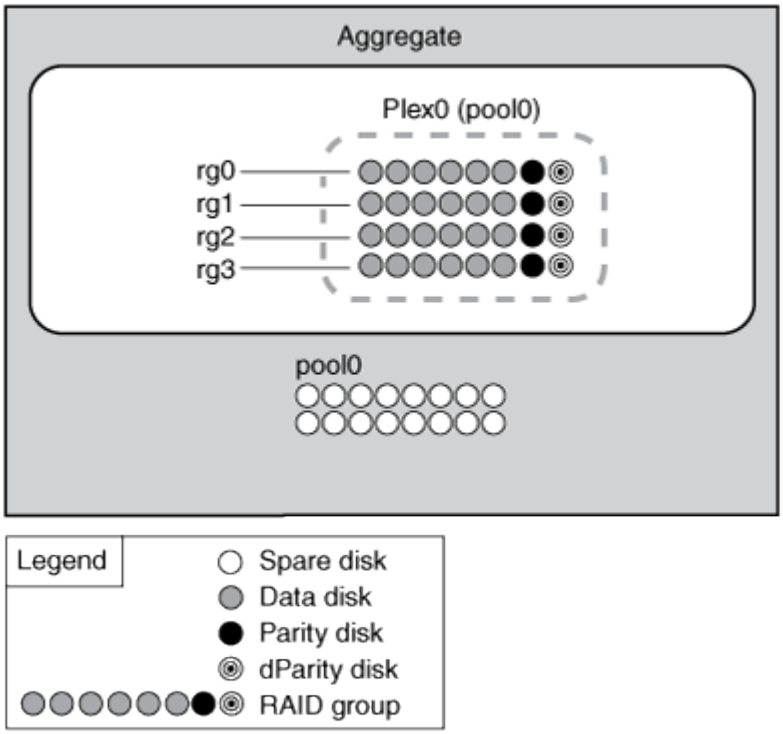
已镜像和未镜像聚合

您可以使用名为 *raid* 的可选功能同步镜像存储在不同 SyncMirror 组中的副本或 *plexes* 中的聚合数据。丛可确保在出现故障的磁盘数量超过 RAID 类型所能保护的磁盘数量或与 RAID 组磁盘的连接断开时防止数据丢失。

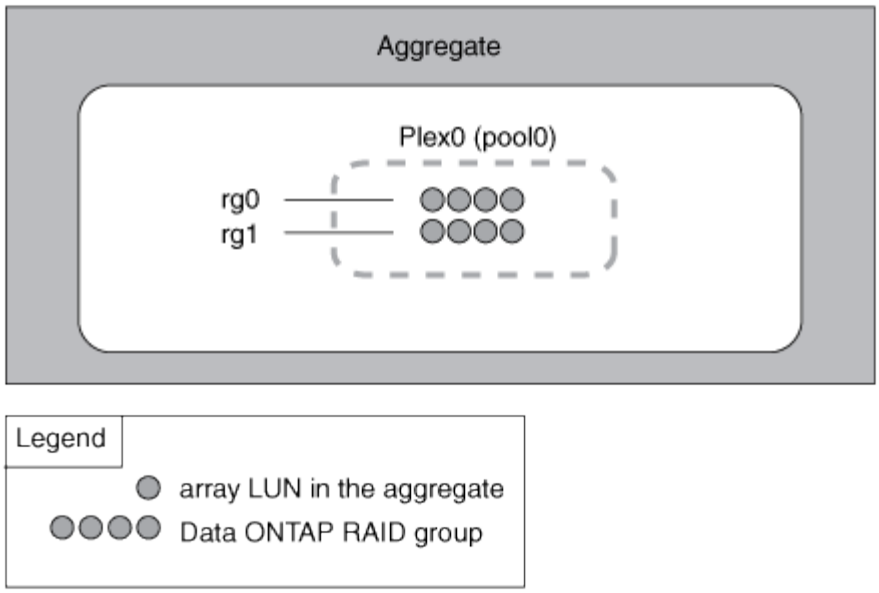
未镜像聚合的工作原理

除非您使用的是 SyncMirror ，否则所有聚合都会取消镜像。未镜像聚合只有一个 *plex* （其数据的副本），其中包含属于该聚合的所有 RAID 组。

下图显示了一个由磁盘组成的未镜像聚合及其一个丛。聚合包含四个 RAID 组： rg0 ， rg1 ， rg2 和 rg3 。每个 RAID 组都有 6 个数据磁盘，一个奇偶校验磁盘和一个双奇偶校验磁盘。聚合使用的所有磁盘都来自同一个池 pool0 。



下图显示了一个包含阵列 LUN 的未镜像聚合及其一个丛。它具有两个 RAID 组： rg0 和 rg1 。聚合使用的所有阵列 LUN 都来自同一个池 pool0 。



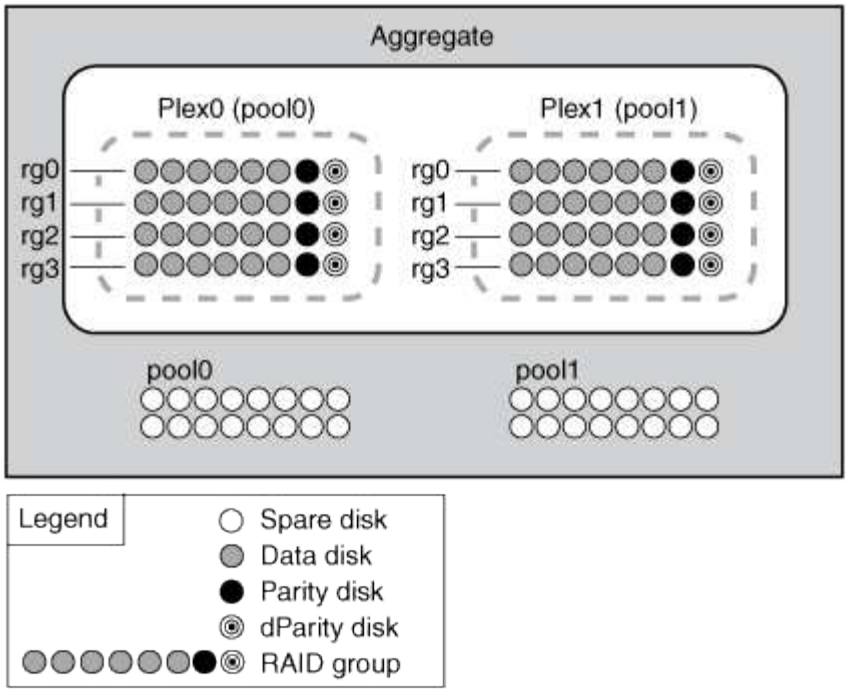
镜像聚合的工作原理

镜像聚合具有两个 *plexes* （其数据的副本），这两个聚合使用 SyncMirror 功能复制数据以提供冗余。

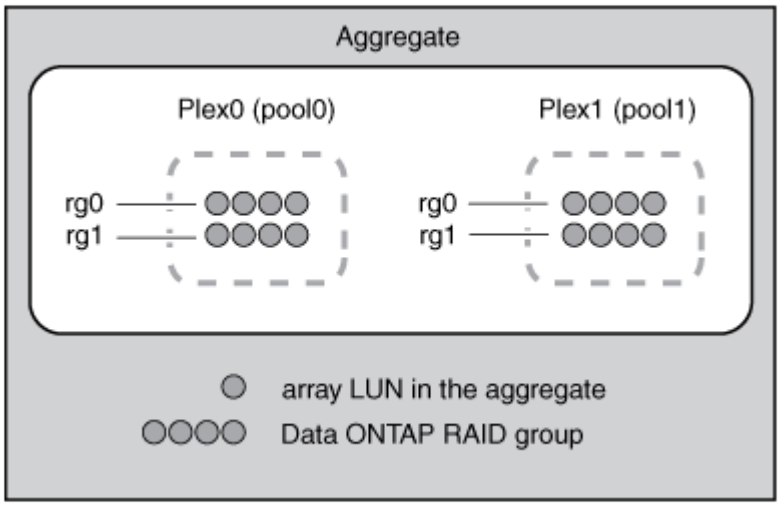
创建镜像聚合（或将第二个丛添加到现有未镜像聚合）时， ONTAP 会将原始丛（ plex0 ）中的数据复制到新丛（ plex1 ）中。丛在物理上是分开的（每个丛都有自己的 RAID 组和池），并且丛会同时更新。这样，如果出现故障的磁盘数超过聚合所保护的 RAID 级别或连接断开，则可以提供额外的保护，防止数据丢失，因为在修复故障发生原因时，未受影响的丛会继续提供数据。修复出现问题的丛后，两个丛将重新同步并重新建立镜像关系。

系统上的磁盘和阵列 LUN 分为两个池： pool0 和 pool1 。Plex0 从 pool0 获取其存储，而 Plex1 从 pool1 获取其存储。

下图显示了由启用和实施了 SyncMirror 功能的磁盘组成的聚合。已为聚合创建第二个丛，即 plex1 。plex1 中的数据是 plex0 中的数据副本， RAID 组也是相同的。32 个备用磁盘分配给 pool0 或 pool1 ，每个池有 16 个磁盘。



下图显示了由启用和实施了 SyncMirror 功能的阵列 LUN 组成的聚合。已为聚合创建第二个丛，即 plex1 。Plex1 是 plex0 的副本， RAID 组也相同。



Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.